

Polyphosphate gel/methyl orange supramolecular composites

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Abstract

The aims of this work were to investigate theoretically the optical properties of methyl orange (MO) and the synthesis of new supramolecular composites based on the incorporation of this dye in an aluminum polyphosphate gel network. The theoretical methodology was based in semiempirical (AM1 and INDO/S-CI) and ab initio (3-21G*) methods. Our results reveal the existence of different electronic patterns for the acidic and basic forms of these molecules. Also, we present a theoretical spectroscopic study for the molecules including interactions with water molecules. MO was successfully incorporated in its acidic form within the host matrix, leading to pink–red transparent self-standing films. The dye could be converted to its basic form upon exposure to ammonia vapor. The spectrum of MO basic form within the gel network differs from its behavior in aqueous solution.

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1. Introduction

The low processing temperature of the sol–gel technique provides a very suitable path to new host–guest structures. A large number of organic and inorganic additives may be incorporated within the matrix network in order to develop materials for new optical devices such as non-linear optical (NLO) materials [1], sensors [2], solid-state lasers [3] and light emitting devices [4].

These materials are mostly silica based glasses [5,6] and hybrid organo-inorganic networks, [7,8]. Rare-earth doped alumina films prepared by the sol–gel technique were recently reported [9].

Polyphosphate-based materials can be synthesized with many different morphologies: rigid foams [10], particles with closed pores [11], nanoparticles [12], thermoreversible gels [13] and hybrid networks [14]. These materials may be prepared in a broad range of chemical compositions, at or near room temperature.

Aluminum polyphosphate (APP) gels, specifically, are supramolecular ionic swollen networks where the polyphosphate chains are interconnected by aluminum ions. Polyphosphate chains

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are linear, short and stiff polyanions, which provide many binding sites with positively charged species could interact. An appropriate control of experimental parameters like P/Al ratio, temperature, acidity and aging/drying allows for the preparation of fibers, flexible or rigid thick films and monolithic bulk pieces at room temperature [13].

Also, these gels are attractive hosts for new hybrid materials. The entrapment of temperature sensitive compounds on the polyphosphate network under mild reaction conditions, is envisaged, thus preventing degradation of the active species. This strategy may lead to new sensors and optical devices. Recently, we reported for first time the synthesis of polyphosphate-based supramolecular composites, with the entrapment of rare-earth cryptates within this matrix [15].

In this work, we describe a new supramolecular composite resulting from the incorporation of methyl orange (MO) within this matrix. MO is a dye widely used as a pH indicator, which also presents NLO properties [16].

2. Experimental

2.1. Theoretical methodology

The semiempirical and *ab initio* geometries were fully optimized for MO molecules (Fig. 1). Acidic and basic MO forms were used. In the basic form, a hydrogen ion is lost from the –NN– bridge between the rings, and the electrons, formerly used to bind the hydrogen, neutralize the positive charge on the terminal nitrogen, so that it is no longer able to π -bond.

The standard 3-21G* basis set was used for *ab initio* calculations of the molecules from Fig. 1. The semiempirical geometric data utilized in this

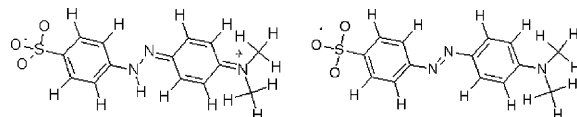


Fig. 1. Schematic chemical structure of methyl orange in the acidic and basic forms, respectively.

paper were obtained from fully optimized by AM1 calculations [17]. For the *ab initio* and semiempirical calculations we have used the Gaussian 94 [18] and the MOPAC [19] packages, respectively. For geometry optimization we performed semiempirical calculations changing the standard convergence criteria adopted on MOPAC to a maximum step size of the 0.01.

To obtain the absorption spectra we have utilized the INDO/S-CI (intermediate neglect of differential overlap/spectroscopic-configuration interaction) method [20–26], because this method was specifically designed to describe the optical-UV–visible transitions of organic compounds [27–31]. We utilized the gammas Mataga-Nishimoto parameters in the theoretical spectroscopy calculations. For these calculations approximately 200 configurations were investigated for each molecule including singlet and doublet states.

For the solvent interaction we utilized a periodic box with $8 \times 8 \times 18 \text{ \AA}^3$ (Fig. 2), including water molecules with a typical water–MO distance of 2.3 Å. The periodic box supported by our calculation contained 16 water molecules.

2.2. Experimental procedure

The incorporation of MO within the APP gel network was carried out starting from sodium polyphosphate (2.0 M, in P), aluminum nitrate (1.0 M) and MO (1.0%) aqueous solutions, which

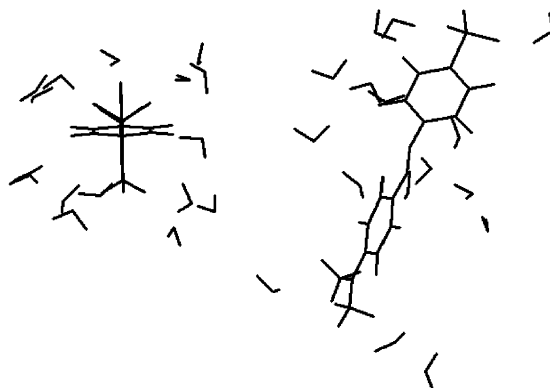


Fig. 2. Representation of MO with water molecules in a virtual periodic box in front view and lateral view, respectively.

were added under stirring. In some cases, sodium hydroxide solution was also added to control the pH. The gel preparation method is described on the literature [13]. The phosphorus to aluminum ratio (P/Al) ranged from 2.0 to 2.8.

Gelation occurs under ambient atmosphere, at room temperature and is followed by syneresis, which may be accelerated by centrifugation cycles (30 min, at 10 000 g). Transparent pink–red samples form after 1–10 days at room temperature. Self-standing films were prepared spreading the gel samples on a flat surface and letting them dry at room temperature in a moisture-controlled environment, for one week.

The samples were characterized by absorption spectra using a UV–visible spectrophotometer Perkin–Elmer LAMBDA 6 model 2688-002, within the 200–800 nm range.

3. Results and discussion

APP gels are colorless and transparent. Hence, dye incorporation within the matrix was initially confirmed by the resulting transparent red–pink samples. Fig. 3 shows the absorption spectra of single matrix and gel/dye composites. The spectra of the as-prepared red–pink samples (Fig. 3(b), solid line) showed the typical transition ascribed to MO in its acidic form in aqueous media, with a maximum located near 510 nm. The gelation occurs at pH values ranging from 0.5 to 2.0, which provides the explanation for this observation.

Upon exposure to ammonia vapor (6 M solution), some samples turned to yellow after 1–4 h. When the exposure is stopped and the samples stored overnight, they acquire a pale yellow coloration. This result indicates that the dye can be converted to its basic form after incorporation within the gel network, as a consequence of two features: (i) the gel is permeable to the vapor; (ii) diffusion processes may occur within the matrix.

Fig. 3(c) (solid line) shows the spectrum of gel/dye sample in which MO is in its basic form (yellow sample), with a maximum at 400 nm. The spectrum differs from what is observed in MO aqueous solution, which is orange and presents an absorbance centered at 465 nm.

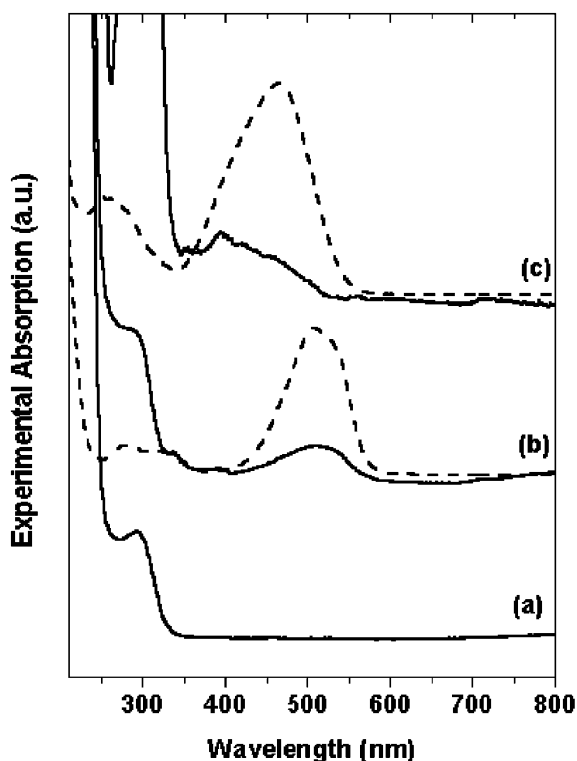


Fig. 3. Experimental absorption spectra. (a) APP gel; (b) MO/APP composite (solid line); MO acidic form in aqueous solution (dashed line); (c) MO/APP composite after exposure to ammonia vapor (solid line); MO basic form (dashed line).

The APP gel is a swollen network that contains nearly 40% of water. When entrapped on this network, the MO molecules must be in an aqueous environment which is more restrictive to the guest molecule as compared with a true solution. Since the conversion to the basic form occurs after gel preparation, this hypothesis could explain the observed difference, which resembles solvatochromism, a well-known phenomenon observed for dye molecules in solution.

In Fig. 4 we have the theoretical absorption spectra for MO acidic and basic forms. The solid line represents the MO molecule without water interaction and the dashed line represents it with water interaction following the steps discussed under theoretical methodology.

For the MO basic molecule the strongest band is at ≈ 385 nm and has an oscillator strength (o.s.) equal to 1.47. This corresponds to a mixing

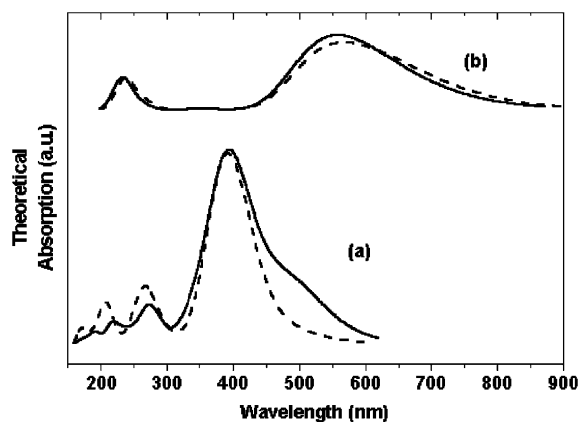


Fig. 4. Theoretical UV-visible spectra for MO in (a) basic form and (b) acidic form with (dashed line) and without (solid line) water interaction. The methodology utilized were INDO/S-CI coupled with AM1 geometries. These spectra were obtained by describing each of the principal transitions using a suitably weighted Gaussian function normalized to the calculated oscillator strengths.

of the transitions $|H-2 \rightarrow L\rangle$, $|H-4 \rightarrow L\rangle$ and $|H-5 \rightarrow L\rangle$ (with coefficients of -0.37 , -0.36 and 0.34 , respectively). Note that, H represents the highest occupied and L the lowest unoccupied molecular orbital. The band at ≈ 270 nm has an o.s. equal to 0.13 and is essentially a pure $|H \rightarrow L+4\rangle$ transition, while the band at ≈ 215 nm corresponds to a mixing of the transitions $|H-3 \rightarrow L+4\rangle$ and $|H-5 \rightarrow L+4\rangle$ (with equal coefficients of 0.4) with o.s. = 0.16 .

Similar behavior is followed by the MO basic molecule with water interaction included (Fig. 4(a), dashed line). The band at ≈ 392 nm has an o.s. equal to 1.34 and corresponds to a mixing of the transitions $|H-4 \rightarrow L\rangle$ and $|H-5 \rightarrow L\rangle$

(with coefficients of -0.38 and 0.50 , respectively). Note that the $H-2$ energy level is not included in the main transitions, which shows the importance of deeper orbitals. The band at ≈ 265 nm is essentially the same transition of the MO basic form (vacuum), but the weight of this transition corresponds to 0.51 (o.s. = 0.23). The band centered at ≈ 210 nm corresponds to a mixing of the transitions $-0.38|H-1 \rightarrow L+5\rangle$ and $-0.46|H-2 \rightarrow L+5\rangle$ with o.s. = 0.19 . Similar analysis was done as shown in Table 1.

A comparison of the theoretical and experimental spectra shows fairly good agreement for dye molecules in acidic form, with a 45 nm displacement to higher wavelength in the calculated spectrum.

Regarding to MO basic form, is important to point out that the maximum of the band in the visible region for the calculated and the gel/dye spectra almost coincide. Since, the calculated spectra does not account for translational motion of the molecules in solution and molecular vibrations, this observation could be evidence that the guest MO species are rather entrapped on the host matrix. This interesting feature certainly deserves further investigation to be confirmed.

4. Conclusions

Acidic and basic MO have been simulated by semiempirical methods. The features in the absorption spectra of the MO molecules including the interaction with water that we report consist of mixing of different transitions and are in good agreement with experiment. Our results show an

Table 1

Main electronic transitions as determined from INDO/S-CI calculations based on optimized ground-state geometry (H , L , $W.L.$ and $O.S.$ represent HOMO, LUMO, wavelength and oscillator strength, respectively)

Molecule	W.L. (nm)	O.S.	Transition (absorption)
MO acidic	555	0.48	$-0.70 H-4 \rightarrow L\rangle$
	232	0.10	$-0.69 H-4 \rightarrow L+4\rangle$
MO acidic + water	573	0.42	$-0.70 H-4 \rightarrow L\rangle$
			$0.60 H \rightarrow L\rangle$
	350	0.01	$0.66 H \rightarrow L+3\rangle$
	237	0.10	$0.69 H-4 \rightarrow L+4\rangle$

extensive conjugation in the present system, the molecule–water has a medium distance of 2.3 Å.

In addition, Al-polyphosphate gel/MO composites were successfully prepared under mild reaction conditions. The dye molecules are solvated by water molecules, which in turn are embedded within a solid matrix, which results from electrostatic interactions between polyphosphate chains and Al^{3+} ions, in a supramolecular array. MO/APP samples are sensitive to acidic and basic vapors, that lead to color change of these samples.

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